Problem 1.4

(a) Fix a natural number $n \in \mathbb{N}$, then the number of n-digit numbers in base r not containing the digit i is exactly $(r-1)^n$. Call this set of numbers $A^{(n)}$ then the minimum distance of the set $A^{(n)}$ to any element not containing i in its base r expansion is r^{-n}. Now simply denote by $B[x, r^{-n}]$ the closed ball around x with radius r^{-n}, then by the argument above one has

$$A_r(i) = \bigcup_{x \in A^{(n)}} B[x, r^{-n}]$$

and

$$\sum_{x \in A^{(n)}} |B[x, r^{-n}]| \leq \left(\frac{r-1}{r}\right)^n \to 0\,, \text{ as } n \to \infty\,,$$

which yields the claim.
Remark: We used terminating sequences to define the set $A^{(n)}$, namely n-digit numbers in base-root (i.e. the expansion in base r ends in a sequence of zeros), although the text omits these to have unique representations. However, for the definition of a *trifling* set this does not matter.

(b) The Cantor set C is an example.